

GFMixer: Decoupled Temporal Gradient and Fourier-Aware Attention for Time Series Forecasting

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Abstract

Multivariate time series forecasting is fundamental to web-scale systems. However, frequency-domain forecasters face two structural challenges: (i) **frequency bias**, where low-amplitude yet informative temporal cues are often submerged in noise and neglected during training; (ii) **spectral degradation**, where standard neural transformations distort high-amplitude periodic structures, thereby weakening the predictive signal. To address both issues, we propose **GFMixer**, a decoupled dual-path architecture. GFMixer first applies a Temporal Gradient Block (TGB) to capture low-amplitude information through adaptive selection based on temporal-geometric evidence. Subsequently, a Fourier-Aware Attention Block (FAB) represents high-amplitude, multi-frequency information to mitigate spectral degradation. Finally, both information streams are integrated via time-aligned residual connections within a Gradient Aggregation Block (GAB) for the final forecasting task. Extensive evaluations across seven standard benchmarks (ETT, Weather, Electricity, and Traffic) demonstrate GFMixer’s superiority, securing 37 first-place rankings and the lowest average MSE overall. Beyond standalone performance, GFMixer serves as a versatile plug-and-play module that consistently enhances mainstream backbones. Code is available at: <https://github.com/superlin30/GFMixer>.

CCS Concepts

• **Mathematics of computing** → **Time series analysis**.

Keywords

Time Series Forecasting, Frequency Bias, Spectral Degradation

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1 Introduction

As the Web operates increasingly as a real-time, data-driven infrastructure, time series forecasting has become a core capability for forecasting user demand and system dynamics, spanning energy management [23], weather prediction [7], and finance [2, 8, 19]. While traditional models focus on temporal dependencies, an increasing amount of research highlights that key predictive structures are often more clearly revealed in the frequency domain [3, 10, 24]. Real-world series typically consist of (i) low-frequency, high-amplitude components that capture long-term or seasonal regularities, and (ii) high-frequency, low-amplitude fluctuations that encode rapid trend reversals and fine-grained accelerations [11]. These weak signals may be small in energy, but they often carry early-warning information that is vital for timely decision-making [5].

However, current frequency-domain modeling pipelines still face two unresolved challenges: The first challenge is frequency bias toward high amplitude components [12]. Mainstream frequency-domain models tend to amplify dominant peaks and suppress low-amplitude components (e.g., top- k selection), causing models to ignore weak but meaningful predictive cues [17]. As shown in Figure 1(a), low-amplitude components often encode high-frequency key temporal patterns such as rapid trend reversals and fine-grained acceleration changes [11]. Prior attempts to “bring back” weak components either equalize attention across bands [12] or amplify low-energy components [5]. Both “overlooking” and “boosting” methods face a fundamental trade-off between signal recovery and stability. Low-amplitude regions mix informative signal with noise, thus equal treatment fragments attention, while indiscriminate amplification boosts noise and induces instability.

The second challenge is spectral degradation. Theoretically, we have proven that, when high-amplitude periodic components are transformed by neural encoders, nonlinear operations (e.g., additive positional encodings and projections) often destroy spectral coherence and blur periodic cues, undermining long-term generalization. As shown in Figure 1(b), these nonlinear operations inherently lack frequency awareness or inductive bias toward periodicity. They inject positional noise, mix spectral bases, and introduce harmonic distortions [6]. Empirically, we demonstrate that periodic protection significantly improves generalization and stability across benchmarks in energy, weather, and financial forecasting tasks.

Even after decoupling low-amplitude cues to mitigate frequency bias and preserving high-amplitude periodicity against spectral degradation, truly separating meaningful cues from noise still requires a shared time coordinate with stable, recurring windows.

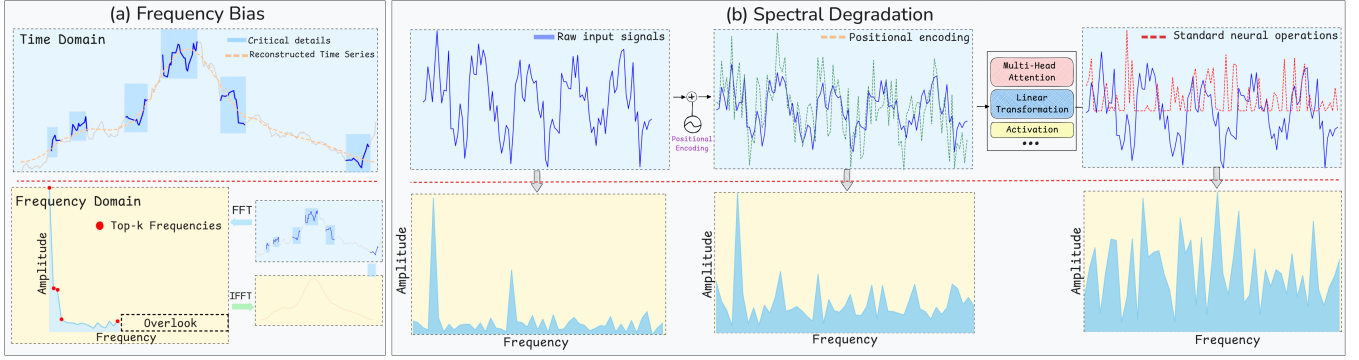


Figure 1: (a) illustrates that top- k frequency selection overlooks low-amplitude components, which leads to frequency bias. We visualize the key pattern loss in the time domain; (b) simulates and visualizes spectral degradation from the time domain and its corresponding frequency domain, illustrating how positional encoding and neural transformations disrupt periodic components.

High-frequency fluctuations comprise two types: (a) random, unanchored impulses that occur idiosyncratically along the time axis, and (b) recurrent, window-specific transients whose probability or intensity systematically varies across within-cycle time slots (e.g., evening load spikes, weekday commute peaks, market open/close windows). Only (b) recurs relative to these windows and thus carries stable predictive value. When temporal alignment is lost, the coordinate that distinguishes (b) from (a) collapses; recurrent transients can no longer bind to their windows and become statistically indistinguishable from noise.

Therefore, a credible time series forecasting framework must jointly treat these two complementary carriers along the shared time axis: (i) preserving backbone periodicity (amplitude and time-axis information) across layers, and (ii) modeling low-amplitude cues decoupled from the backbone stream and injecting them selectively into the backbone along the shared time axis. Theoretically, we formalize the above challenges and prove several structural results that motivate our decoupled design. Our approach, **GFMixer**, implements our principle with a dual-path architecture. As illustrated in Figure 2, GFMixer consists of three main components:

First, we introduce a Temporal Gradient Block (TGB) to capture weak yet predictive local trends and turning points that are commonly overlooked by frequency-domain forecasters. TGB extracts low-amplitude components from the signal by computing multi-scale causal geometric features, specifically temporal slopes and curvatures, which highlight local variations that become informative only when aligned to a unified time axis. TGB operates in parallel with the periodic backbone to avoid distorting dominant frequency components and injects gradient information through a two-level gating mechanism. A task-adaptive gate controls the relative importance of slope-based versus curvature-based evidence, enabling the model to adjust to different dataset characteristics and training stages. A temporal-feature gate further modulates these signals across time steps and variables using recurring temporal patterns inferred from the backbone’s seasonal and positional encodings, amplifying contributions during historically relevant periods and suppressing them elsewhere.

Second, we introduce a Fourier-Aware Attention Block (FAB) that embeds positional information into a multi-frequency Fourier representation, enabling the attention mechanism to preserve spectral integrity and mitigate the spectral degradation commonly induced by deep neural transformations.

Finally, we design a Gradient Aggregation Block (GAB) to refine the backbone output through a residual correction pathway. GAB enhances low-amplitude temporal variations without disturbing the high-amplitude structures learned by the backbone by injecting fused gradient signals that capture geometric dynamics into the final prediction along the unified time axis.

Importantly, this gradient aggregation is structurally decoupled from the backbone, enabling stable and plug-and-play integration of TGB and FAB with diverse forecasting architectures. The unique contributions of our study are threefold:

- We identify and formalize two fundamental challenges in frequency-domain forecasting: **frequency bias**, which hinders the modeling of low-amplitude yet informative components, and **spectral degradation**, where high-amplitude periodic structures are distorted by neural transformations. We show how positional encodings, projections, and nonlinearities induce cross-frequency mixing that exacerbates the issue.
- We propose **GFMixer**, a decoupled dual-path framework comprising (i) a Temporal Gradient Block (TGB) with task-adaptive and temporal-feature gating to extract weak geometric signals, (ii) a Fourier-Aware Attention Block (FAB) to preserve multi-frequency periodic structures, and (iii) a Gradient Aggregation Block (GAB) for residual, window-aware correction.
- Extensive experiments on seven benchmarks (ETT, Weather, Electricity, and Traffic) show that GFMixer achieves **37 best results** and the **lowest average MSE**. It also serves as a lightweight plug-and-play module that consistently improves mainstream backbones. Code is available at <https://github.com/superlin30/GFMixer>.

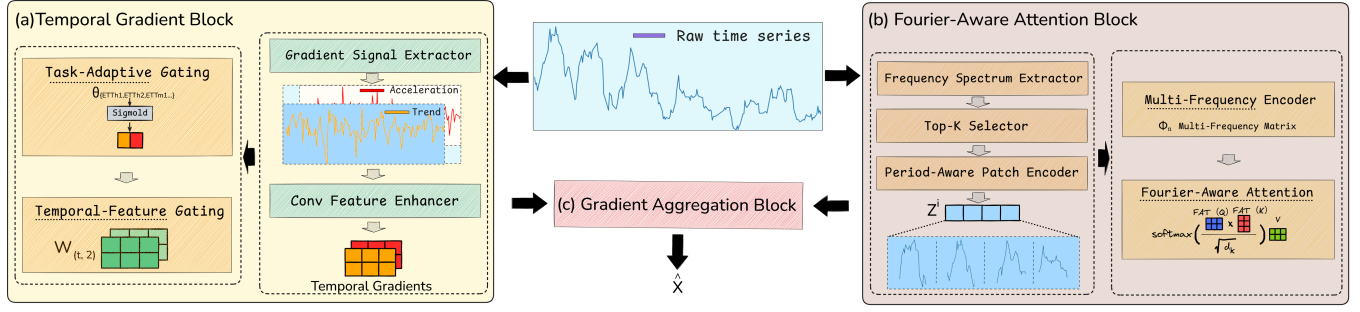


Figure 2: Overview of the proposed GFMixer framework for time series forecasting.

2 Related Work

2.1 Decomposition and Frequency-Domain Modeling

Time series often exhibit periodic patterns, such as daily rhythms and seasonal cycles. Classical decomposition-based models, including DLinear [22], Autoformer [18], and FEDformer [25], explicitly separate time series into trend and seasonal components, applying attention mechanisms tailored to each. While effective in capturing coarse temporal structures, these models remain confined to the time domain. Recently, frequency-domain approaches have gained traction for their ability to reveal latent periodicity. TimesNet [17] and Times2D [9] utilize Fourier transforms to convert time series into 2D frequency representations, enabling convolutional modules to capture both intra- and inter-period patterns. Similarly, PDF [4] segments sequences into frequency bands, explicitly modeling short- and long-term dynamics in separate spectral partitions. However, attention-based architectures, despite their success in sequential modeling, are not inherently frequency-aware. As deep transformations such as positional encodings and nonlinear activations are applied, frequency structures tend to degrade, undermining the preservation of periodic signals. In contrast, GFMixer introduces a Fourier-Aware Attention Block (FAB) that explicitly embeds multi-frequency position representations into the attention mechanism itself, preserving spectral integrity throughout the modeling process.

2.2 Low-Amplitude Components Modeling

Most frequency-domain forecasting models emphasize the dominant, high-energy components, such as the largest amplitude or the top periodic modes, while often overlooking low-amplitude patterns that may encode subtle but critical temporal patterns. To mitigate this bias, Fredformer [12] equalizes attention across spectral bands, while Amplifier [5] boosts low-energy components via energy amplification modules. Despite their innovations, these approaches present limitations. Fredformer is tightly coupled to Transformer architectures and Amplifier, while generalizable, amplifies all low-energy content, lacking a method to discriminate noise from informative signals. GFMixer adopts a fundamentally different strategy: it structurally decouples low-amplitude modeling from the main network via a Temporal Gradient Block (TGB). This block computes temporal gradients, capturing both trend and

acceleration signals, and employs a two-level weighting mechanism to selectively fuse informative key patterns while preventing potential noise from contaminating the core backbone. This design enables stable, context-aware enhancement of weak signals.

3 Preliminaries

3.1 Frequency Bias

Notation and assumptions. We consider a real sequence $x = p + r + \varepsilon \in \mathbb{R}^N$. Let $F \in \mathbb{C}^{N \times N}$ be the *unitary* DFT so that $F^*F = I$ and $\hat{u} \triangleq Fu$. Write $P_k \triangleq F^* \Pi_{\Omega_k} F$, where Π_{Ω_k} projects onto the index set $\Omega_k \subset \{0, \dots, N-1\}$ of k kept frequencies. Unless stated otherwise, Ω_k is obtained by Top- k selection on the magnitude spectrum of the periodic backbone p : $\Omega_k = \text{TopK}(\{|\hat{p}_\omega|\}_\omega)$, with ties broken by increasing index.¹ We model transients by the class

$$\mathcal{R}(s, \Omega, \eta) \triangleq \left\{ r \in \mathbb{R}^N : r \text{ is } s\text{-sparse in time, } |\text{supp}(\hat{r})| \geq \Omega, \right. \\ \left. \max_{\omega \in \text{supp}(\hat{r})} |\hat{r}_\omega| \leq \eta \|r\|_2 / \sqrt{\Omega} \right\}.$$

Here $\Omega \in \{1, \dots, N\}$ is a *lower bound* on the Fourier support size of r (not to be confused with Ω_k), and $\eta > 0$ controls its peak-to-energy ratio. Throughout, expectations are with respect to the data-generating distribution and all random variables are square-integrable.

Unavoidable loss. For any k -line truncation and $r \in \mathcal{R}(s, \Omega, \eta)$,

$$\|P_k r\|_2^2 \leq \eta^2 \frac{k}{\Omega} \|r\|_2^2 \implies \|(I - P_k)r\|_2^2 \geq (1 - \eta^2 \frac{k}{\Omega}) \|r\|_2^2. \quad (1)$$

Prediction risk as a minimization gap. Assume $y = \langle \beta, x \rangle + \xi$ with $\beta \in \mathbb{R}^N$, and ξ zero-mean noise independent of x with $\mathbb{E}\xi^2 < \infty$. Let $Z = P_k x$. We make the hypothesis spaces explicit:

$$\mathcal{F}_Z := \{\phi : \mathbb{R}^N \rightarrow \mathbb{R} \text{ Borel-measurable, } \mathbb{E}[\phi(Z)^2] < \infty\},$$

$$\mathcal{F}_x := \{\psi : \mathbb{R}^N \rightarrow \mathbb{R} \text{ Borel-measurable, } \mathbb{E}[\psi(x)^2] < \infty\}.$$

Then the exact excess Bayes risk of observing Z in lieu of x is

$$\begin{aligned} \text{ExcessMSE} &\triangleq \inf_{\phi \in \mathcal{F}_Z} \mathbb{E}[(y - \phi(Z))^2] - \inf_{\psi \in \mathcal{F}_x} \mathbb{E}[(y - \psi(x))^2] \\ &= \mathbb{E}[\text{Var}(\langle \beta_\perp, (I - P_k)x \rangle \mid Z)], \end{aligned} \quad (2)$$

¹All results hold for any deterministic, tie-broken size- k set Ω_k .

where $\beta_\perp \triangleq (I - P_k)\beta$. Under a mild alignment/no-leakage condition stated in the Appendix, let $\alpha \triangleq \langle \beta_\perp, r \rangle / (\|\beta_\perp\|_2 \|r\|_2) \in [-1, 1]$. Then

$$\begin{aligned} \text{ExcessMSE} &\geq \|\beta_\perp\|_2^2 \alpha^2 \mathbb{E} \left[\frac{\|(I - P_k)r\|_2^4}{\|r\|_2^2} \right] \\ &\geq \|\beta_\perp\|_2^2 \alpha^2 \left(1 - \eta^2 \frac{k}{\Omega}\right)^2 \mathbb{E} \|r\|_2^2, \end{aligned} \quad (3)$$

where (here and below) k is the number of kept frequencies in P_k , Ω and η are as defined above for $r \in \mathcal{R}(s, \Omega, \eta)$.

Instability of forcing low-energy bands. Consider amplification in the same pipeline via $A_w = F^* \text{diag}(w) F$. Let the *weak-band index* set be $\mathcal{S} \subseteq \{0, \dots, N-1\}$ (typically $\mathcal{S} = \Omega_k^c$), and define

$$w_\omega = \begin{cases} \gamma > 1, & \omega \in \mathcal{S}, \\ 1, & \omega \notin \mathcal{S}. \end{cases}$$

With unitary normalization of F , we have $\|A_w\|_2 = \max_\omega |w_\omega| \geq \gamma$ and $\kappa(A_w) = \frac{\max_\omega |w_\omega|}{\min_\omega |w_\omega|} \geq \gamma$. For white noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $\mathbb{E} \|A_w \varepsilon\|_2^2 = \sigma^2 \sum_\omega |w_\omega|^2 \geq \sigma^2 |\mathcal{S}| \gamma^2$. If we enforce the stability constraint $\|A_w\|_2 \leq C$, then for any k and any transient $r \in \mathcal{R}(s, \Omega, \eta)$,

$$\|P_k A_w r\|_2^2 \leq C^2 \eta^2 \frac{k}{\Omega} \|r\|_2^2, \quad (4)$$

so the recoverable transient energy remains $\tilde{O}(k/\Omega)$.

Eqs. (1)–(4) together show a structural truncation loss and a stability–gain tension, motivating our *decoupled* architecture. The full proofs are in Appendix A.1.

3.2 Spectral Degradation in Deep Models

We quantify how standard neural operations degrade periodic structure at the patch level. Fix a selected angular frequency $\omega_i \in (0, \pi]$, patch length $p_i = \text{round}(2\pi/\omega_i)$ with consistent zero-padding, and intra-patch unitary DFT F_{p_i} . Let $D \triangleq F_{p_i} \otimes I_N$ and Π_{p_i} be the orthogonal projector onto the p_i -periodic subspace.² For any patch $u \in \mathbb{R}^{p_i \times N}$, the *best periodic fit* solves

$$\min_{v \in \mathcal{V}_{p_i}} \|u - v\|_F^2 = \|\Pi_{p_i}^\perp u\|_F^2, \quad \mathcal{V}_{p_i} \triangleq \{v : v \text{ is } p_i\text{-periodic}\}. \quad (5)$$

We study one layer that acts on a patch z as

$$z \xrightarrow{\text{APE}} z + P \xrightarrow{W} W(z + P) \xrightarrow{\phi} \phi(W(z + P)),$$

where P is an additive positional encoding on the patch, W a linear map, and ϕ a pointwise nonlinearity.

LEMMA 3.1 (EXACT DECOMPOSITION FOR APE). *For any z, P ,*

$$\begin{aligned} \|\Pi_{p_i}^\perp(z + P)\|_F^2 &= \|\Pi_{p_i}^\perp z\|_F^2 + \|\Pi_{p_i}^\perp P\|_F^2 \\ &\quad + 2\langle \Pi_{p_i}^\perp z, \Pi_{p_i}^\perp P \rangle_F. \end{aligned}$$

In particular, if P is non- p_i -periodic or varies with the patch index, it introduces a nonperiodic component that cannot be removed by projecting onto $\mathcal{V}_{p_i}$.

THEOREM 3.2 (LINEAR FREQUENCY PRESERVATION $\Leftrightarrow$ COMMUTATION). *The following are equivalent: (i) for every intra-patch DFT bin ℓ , Dz supported on bin ℓ implies $DWD^{-1}Dz$ is supported on bin ℓ (no mixing); (ii) $(S \otimes I_N)W = W(S \otimes I_N)$, where S is the cyclic shift on $\mathbb{R}^{p_i}$; (iii) $W = D^{-1}\tilde{W}D$ with $\tilde{W}$ block-diagonal by bins. If*

²DC ($\omega = 0$) is handled separately and is never mapped to a finite period.

$[W, D] \triangleq WD - DW \neq 0$, there exists a single-bin input whose output leaks energy outside that bin, with leakage $\gtrsim \|[W, D]\|_2$.

THEOREM 3.3 (NONLINEAR HARMONICS). *Let $\phi(u) = \sum_{m \geq 0} a_m u^m$ be analytic (piecewise-linear cases follow by local polynomialization). For $x(\tau) = \sum_{j=1}^J A_j \cos(\omega_j \tau + \varphi_j)$, the DFT of $\phi(x)$ has support on Minkowski sums $\sum_j s_j \omega_j$ ($s_j \in \mathbb{Z}$) modulo 2π , with nonzero coefficients whenever some $a_m \neq 0$ for $m \geq 2$. In particular, for two tones $\omega_1 \neq \omega_2$ and non-affine ϕ ,*

$$\|\Pi_{\{\omega_1, \omega_2\}}^\perp \widehat{\phi(x)}\|_2^2 \geq c_2 |a_2|^2 A_1^2 A_2^2 + c_3 |a_3|^2 (A_1^6 + A_2^6) + \dots,$$

with constants $c_m > 0$ depending only on windowing.

Minimal quantitative guarantee (one layer). Define the per-layer periodic residual

$$\begin{aligned} \text{PerErr}(z; P, W, \phi) &\triangleq \min_{v \in \mathcal{V}_{p_i}} \|\phi(W(z + P)) - v\|_F^2 \\ &= \|\Pi_{p_i}^\perp \phi(W(z + P))\|_F^2. \end{aligned}$$

Assume (i) $[W, D] \neq 0$ (the layer is not block-circulant on the patch axis) and (ii) ϕ is non-affine (some $a_m \neq 0$, $m \geq 2$). If, in addition, the nonperiodic part of P is conditionally uncorrelated with that of z across patches,³ then there exist universal $c_{\text{lin}}, c_{\text{nl}} > 0$ (depending only on normalization/windowing) such that

$$\begin{aligned} \mathbb{E} \text{PerErr}(z; P, W, \phi) &\geq c_{\text{lin}} \|[W, D]\|_2^2 \mathbb{E} \|z\|_F^2 \\ &\quad + c_{\text{nl}} \sum_{m \geq 2} |a_m|^2 \mathbb{E} [\Theta_m(z, P, W)] - O(p_i^{-1}), \end{aligned} \quad (6)$$

where $\Theta_m(\cdot) \geq 0$ collects the m -th order intermodulation energy produced when ϕ acts on $W(z + P)$. Consequently, the periodic-energy ratio $R(u) \triangleq \|\Pi_{p_i} u\|_F^2 / \|u\|_F^2$ satisfies

$$\begin{aligned} \mathbb{E} R(\phi(W(z + P))) &\leq \mathbb{E} R(z) - \underbrace{\tilde{\rho}_{\text{lin}}}_{\propto \|[W, D]\|_2^2} - \underbrace{\tilde{\rho}_{\text{nl}}}_{\propto \sum_{m \geq 2} |a_m|^2} + O(p_i^{-1}), \end{aligned} \quad (7)$$

even for a single layer, spectral degradation is quantitatively lower bounded by the commutator norm $\|[W, D]\|_2$ (linear mixing) and the nonlinearity order (harmonics/intermodulation). Equality in Eqs. (6)–(7) occurs only if $[W, D] = 0$ (block-circulant/LTI), ϕ is affine, and P is both p_i -periodic and patch-stationary.

The analysis of spectral degradation is not heuristic: Theorem 3.3 formally derives the combination lines (including $\omega_1 \pm \omega_2$) and gives explicit energy lower bounds. We omit these in the main text for brevity and give full proofs in Appendix A.2. In Section 4.2, we introduce the Fourier-Aware Attention mechanism, which mitigates these distortions and preserves the periodic structure, and enhances long-term forecasting performance.

4 GFMixer

In time series forecasting, given a multivariate time series $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times N}$, where T is the sequence length and N is the number of variables, our objective is to predict the next P time steps based on a sliding window of size S . The dataset is segmented into input-output pairs, and for parallel training, we organize them into batches of size B , resulting in $\mathbf{X}_{\text{in}} \in \mathbb{R}^{B \times S \times N}$

³Formally, $\mathbb{E}[\langle \Pi_{p_i}^\perp z, \Pi_{p_i}^\perp P \rangle_F] = 0$ over patches/batches. This mild “no adversarial cancellation” condition removes a purely deterministic cancellation corner case and is standard in practice for learned PEs.

and $\mathbf{X}_{\text{out}} \in \mathbb{R}^{B \times P \times N}$. For simplicity, in the following discussion, we will ignore the batch dimension B and consider the case where the batch size is 1.

4.1 Periodic Representation

Frequency Spectrum Extractor. To model periodicity in time series forecasting, we first extract the frequency-domain patterns through the Fast Fourier Transform (FFT) [1]. For a multivariate time series $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times N}$, the FFT is applied along the temporal axis to obtain the complex frequency coefficients:

$$\mathbf{X}_f = \sum_{t=0}^{T-1} \mathbf{X}(t) \cdot e^{-2\pi i f t / T}, \quad f = 0, \dots, T-1. \quad (8)$$

The magnitude spectrum $\mathbf{A}_f = |\mathbf{X}_f|$ represents the energy distribution across frequencies. This spectrum serves as a frequency-domain inductive prior, aligning well with both prominent periodic structures and subtle temporal patterns.

Top- k Selector. To capture the most dominant periodic patterns, focusing on high-amplitude components, we extract the top- k frequencies with the largest spectral magnitudes. Before top- k selection, spectra are normalized per channel to avoid scale bias. Ties are resolved by prioritizing lower frequencies, and conjugate pairs are treated as a single frequency line. By selecting these frequencies $\{f_1, \dots, f_k\}$ from the magnitude spectrum $\mathbf{A}_f$, we capture the most significant cycles. This procedure is formalized as:

$$\{f_1, \dots, f_k\} = \text{TopK}(\mathbf{A}_f), \quad \text{where } p_i = \left\lceil \frac{T}{f_i} \right\rceil, \quad (9)$$

each selected frequency f_i corresponds to a dominant period p_i , facilitating a concise representation of the signal's long-range structure. Despite the effectiveness of the top- k selection scheme [4, 9, 17], it may overlook low-amplitude yet crucial components, leading to frequency bias [5, 12]. We discuss this problem in the following sections.

Period-Aware Patch Encoder. To further exploit the extracted dominant periodic structure, we segment the original time series into temporally aligned patches. Each selected frequency $f_i \in \mathcal{F}$ corresponds to a characteristic temporal cycle of length p_i . We reshape the sequence $\mathbf{X} \in \mathbb{R}^{T \times N}$ into fixed-length patches of size p_i , yielding a patch token sequence:

$$\mathbf{Z}^{(i)} = [\mathbf{Z}^{(i)}[0], \dots, \mathbf{Z}^{(i)}[L_i - 1]] \in \mathbb{R}^{L_i \times p_i \times N}, \quad (10)$$

where $L_i = \lfloor T/p_i \rfloor$. Each $\mathbf{Z}^{(i)}[n]$ represents a temporal segment spanning one dominant cycle. Under a clean periodic signal, these patches are approximately repeated across time:

$$\mathbf{Z}^{(i)}[n+1] \approx \mathbf{Z}^{(i)}[n], \quad (11)$$

which encodes periodicity implicitly as cross-patch similarity.

4.2 Fourier-Aware Attention Block

To maintain the periodic structure during deep modeling, we introduce the Fourier-Aware Attention Block (FAB). The FAB injects multi-scale frequency alignment into the attention mechanism. The FAB modulates the query and key vectors based on frequency, thereby enhancing their sensitivity to spectral structures while

preserving temporal alignment. Although previous studies have examined multi-frequency modulation in attention for long-sequence modeling [6, 13, 20], these methods are primarily intended for language models and concentrate on encoding positional context. In contrast, we reinterpret frequency injection as a functional modulation of attention specifically tailored to time series forecasting. Our approach directly aligns attention weights with meaningful temporal periodicities. This enables spectral consistency across layers via a Multi-Frequency Encoder based on Fourier-mixed rotational position encoding. This function applies multi-scale frequency modulation to the query q and key k , enhancing their periodicity preservation within the attention mechanism:

Fourier-Aware Attention. We define the Fourier-Aware Attention as:

$$\text{FA-Attention}(q', k', v) = \text{softmax} \left(\frac{q' \cdot k'^T}{\sqrt{d_k}} \right) v, \quad (12)$$

$$q' = \text{FAT}(q), \quad k' = \text{FAT}(k),$$

where $q, k \in \mathbb{R}^{B \times H \times T \times d_k}$ are the query and key tensors after linear transformation, with B as the batch size, H the number of heads, T the sequence length, and d_k the dimensionality per head.

Fourier Alignment Transformation. The Fourier Alignment Transformation (FAT) function applies a rotation-based modulation using multi-frequency positional encodings:

$$\text{FAT}(q) = q \cdot \cos(\Phi) + \text{Rotate}(q) \cdot \sin(\Phi), \quad (13)$$

where $\text{Rotate}(q)$ swaps and negates halves of q , preserving orthogonal sinusoidal structure, and Φ encodes multi-scale frequency projections.

Multi-Frequency Encoder. Unlike standard rotary embeddings with fixed frequencies, we define a learnable frequency basis to inject richer inductive bias. Let $\{\omega_1, \dots, \omega_K\} \subset [\omega_{\min}, \omega_{\max}]$ be non-uniform frequencies initialized from a controlled distribution (e.g., log-spaced or truncated normal). For each position n , the rotation angles are computed as:

$$\Phi_n = [n \cdot \omega_1, \dots, n \cdot \omega_K] \cdot \begin{pmatrix} \mathbf{W}^{\cos} & \mathbf{W}^{\sin} \end{pmatrix}^T, \quad (14)$$

where $\mathbf{W}^{\cos}, \mathbf{W}^{\sin} \in \mathbb{R}^{K \times d_k/2}$ are learnable projection matrices. These projects multiple sinusoidal bases into latent space, enabling each query/key pair to be modulated by a rich spectrum of temporal periodicity.

This formulation can be interpreted as injecting a *sparse spectral envelope* into attention weights, acting as a soft frequency selector. Our method allows the model to learn and align to dominant cycles dynamically, helping maintain spectral coherence across layers and improving extrapolation on periodic signals.

4.3 Temporal Gradient Block

To address the issue of low-amplitude components often being overlooked, we propose a complementary Temporal Gradient Block. This module extracts gradient signals like trend and acceleration signals from the original time series, providing a direct representation of key patterns in low-amplitude components.

Gradient Signal Extractor. Let $\mathbf{X} \in \mathbb{R}^{B \times T \times N}$ represent the input sequence, we calculate the Trend and Acceleration signals as first- and second-order differences:

$$\mathbf{T}(t) = \mathbf{X}(t) - \mathbf{X}(t-1), \quad (15)$$

$$\mathbf{A}(t) = \mathbf{T}(t) - \mathbf{T}(t-1). \quad (16)$$

Conv Feature Enhancer. A convolutional layer is applied to the stacked trend and acceleration signals to enhance meaningful temporal features:

$$\mathbf{TG}(t, m) = \text{Conv}(\text{Stack}(\mathbf{T}(t), \mathbf{A}(t))). \quad (17)$$

Task-Adaptive Gating. To adaptively prioritize trend or acceleration signals across forecasting tasks, a learnable strategy vector θ is used to compute a global weighting, $\alpha_m = \text{Softmax}(\theta)$. This global gating mechanism enables the model to dynamically adjust its focus on trend or acceleration signals depending on the task at hand.

Temporal-Feature Gating. A learnable weight tensor $\mathbf{W}[n, m]$ is introduced, encoding the importance of trend or acceleration for each time step and feature.

Finally, the fused gradient signal is computed as:

$$\mathbf{TG}_{\text{fused}}(t, n) = \sum_{m=1}^2 \alpha_m \cdot \mathbf{W}_{t,n,m} \cdot \mathbf{TG}(t, n, m). \quad (18)$$

4.4 Gradient Aggregation Block

The Gradient Aggregation Block (GAB) serves as the critical bridge that unifies the complementary carriers along the shared time axis. While the FAB directly generates predictions in the forecast horizon P , the gradient signals extracted by the TGB are derived from the historical context window S . We employ a learnable projection via a one-dimensional convolution with kernel size 1 to bridge the temporal mismatch between the historical context window S and the prediction horizon P . Let $\mathbf{TG}_{\text{fused}} \in \mathbb{R}^{S \times N}$ be the aggregated historical gradient features. We employ a temporal projection matrix $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{P \times S}$ to translate these historical dynamics into future corrections, which are residually added to the backbone prediction $\hat{\mathbf{X}}_{\text{base}}$:

$$\hat{\mathbf{X}} = \hat{\mathbf{X}}_{\text{base}} + \mathbf{W}_{\text{proj}} \mathbf{TG}_{\text{fused}}. \quad (19)$$

This mechanism directly maps historical acceleration trends (length S) to the forecast horizon (length P). This design ensures that the correction is strictly aligned with the prediction timeline. The fused signals are added directly without undergoing additional transformations. Furthermore, this design supports lightweight structural modularity that can be directly integrated into existing encoder-based forecasting models. In our experiments, it is seamlessly integrated into PatchTST [10], the FAB replaces standard attention without major architectural changes, and TGB is applied as an external residual path, independent of the backbone model. This design demonstrates the versatility and interoperability of our modules.

5 Experiments

Datasets. We conduct experiments on seven popular real-world datasets [18, 24]: Electricity Transformer Temperature (ETT) with

four sub-datasets (ETTh1, ETTh2, ETTm1, ETTm2), Weather, Electricity, and Traffic. The train/val/test splits follow the protocol established by [24] for the ETT datasets, and the protocol from [17] for the remaining datasets.

Baselines. To comprehensively evaluate the effectiveness of our model, we compare it with a wide range of state-of-the-art models in long-term time series forecasting, categorized as follows:

- **Time-Domain Modeling:** TimeMixer++ [15], FEDformer [25], PatchTST [10], DLinear [22], and CARD [21].
- **Frequency-Domain Modeling:** TimesNet [17], PDF [4], and Times2D [9].
- **Low-Amplitude Components Modeling:** Fredformer [12] and Amplifier [5].

Setups. We use Mean Square Error (MSE) and Mean Absolute Error (MAE) as evaluation metrics, consistent with previous work [16]. All models follow the forecasting setting proposed by TimesNet [17], with a common prediction length $H = \{96, 192, 336, 720\}$. For the look-back window L , we adopt the officially reported configurations or the best-performing settings from official open-source codes. For instance, Times2D and PDF employ $L = 720$, while PatchTST and DLinear utilize $L \in \{336, 512\}$. Our objective is to demonstrate the competitiveness of GFMixer against these advanced models. Experiments were conducted using PyTorch on NVIDIA A100-PCIE-40GB GPUs, following the training and optimization settings of TimesNet [17]: Adam optimizer (initial learning rate 1×10^{-3}), batch size 128, and early stopping with a patience of 10 epochs. We report the mean results averaged over three runs with different random seeds. Statistical tests verify that GFMixer achieves robust and significant improvements ($p < 0.05$) compared to the baselines.

5.1 Main Results

In Table 1, we report multivariate long-term forecasting results across seven benchmark datasets (ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity and Traffic) and four prediction horizons $H \in \{96, 192, 336, 720\}$. GFMixer consistently achieves strong performance and displays remarkable consistency across all datasets and horizons. On average, it obtains the lowest MSE on all seven datasets and the lowest MAE on four out of seven, outperforming state-of-the-art baselines. GFMixer improves MSE by 11.3% and MAE by 8.1% compared to the classic transformer-based baselines (FEDformer and PatchTST), and outperforms linear models like DLinear by over 6.5% in MSE and 5.4% in MAE. Even compared with recently proposed models such as TimeMixer++ (2025), Amplifier (2025) and Times2D (2025), GFMixer achieves comparable performance in certain cases, while achieving advantages in average performance on all datasets. These comprehensive results highlight GFMixer as a new state-of-the-art model for multivariate long-term time series forecasting, capable of maintaining predictive accuracy across various datasets and horizons.

5.2 Ablation Studies

Impact of the Temporal Gradient Block. To evaluate the benefit of incorporating low-amplitude signals via gradients, we compare the full model with three variants: (i) **w/o TGB**, which removes the Temporal Gradient Block entirely; (ii) **w/o Acc**, which retains only the

Table 1: Multivariate long-term forecasting full results with different prediction lengths $H \in \{96, 192, 336, 720\}$. Best results are in bold and second best are underlined.

Models	Metric	H	GFMixer (Ours)		TimeMixer++ (2025)		Amplifier (2025)		Times2D (2025)		PDF (2024)		Fredformer (2024)		CARD (2024)		TimesNet (2023)		DLinear (2023)		PatchTST (2023)		FEDformer (2022)	
			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1		96	0.353	0.390	0.361	0.403	0.371	0.392	0.362	0.395	<u>0.359</u>	0.393	0.373	0.392	0.383	<u>0.391</u>	0.384	0.402	0.375	0.399	0.370	0.399	0.376	0.419
		192	<u>0.391</u>	<u>0.414</u>	0.416	0.441	0.426	0.422	0.405	0.423	0.390	0.413	0.433	0.420	0.435	0.420	0.436	0.429	0.405	0.416	0.413	0.416	0.420	0.448
		336	0.402	0.421	0.430	0.434	0.448	0.434	0.423	0.433	<u>0.420</u>	<u>0.431</u>	0.470	0.437	0.479	0.442	0.491	0.469	0.439	0.443	0.422	0.436	0.459	0.465
		720	0.446	0.463	0.467	0.451	0.476	0.464	0.446	0.467	0.474	0.485	0.467	<u>0.456</u>	0.471	0.461	0.521	0.500	0.472	0.490	0.447	0.466	0.506	0.507
		Avg	0.398	0.422	0.419	0.432	0.430	0.428	<u>0.409</u>	0.430	0.411	0.431	0.435	<u>0.426</u>	0.442	0.429	0.458	0.450	0.422	0.437	0.413	0.429	0.440	0.460
ETTh2		96	0.270	0.332	0.276	0.328	0.279	0.337	0.270	0.334	0.271	0.333	0.293	0.342	0.281	<u>0.330</u>	0.340	0.374	0.289	0.353	0.274	0.336	0.358	0.397
		192	0.335	0.374	0.342	<u>0.379</u>	0.359	0.389	0.337	0.381	<u>0.337</u>	<u>0.379</u>	0.371	0.389	0.363	0.381	0.402	0.414	0.383	0.418	0.339	0.379	0.429	0.439
		336	0.324	0.380	0.346	0.398	0.377	0.406	0.329	0.385	<u>0.325</u>	0.382	0.382	0.409	0.411	0.418	0.452	0.452	0.448	0.465	0.329	0.380	0.496	0.487
		720	0.380	0.425	0.392	0.415	0.420	0.432	0.376	0.422	0.383	0.428	0.415	0.434	0.416	0.431	0.462	0.468	0.605	0.551	<u>0.379</u>	<u>0.422</u>	0.463	0.474
		Avg	0.327	0.378	0.339	0.380	0.359	0.391	<u>0.328</u>	0.381	0.329	0.380	0.365	0.393	0.368	0.390	0.414	0.427	0.431	0.446	0.330	<u>0.379</u>	0.436	0.449
ETTm1		96	0.276	0.338	0.310	0.334	0.316	0.355	0.285	0.340	<u>0.277</u>	<u>0.337</u>	0.326	0.361	0.316	0.347	0.338	0.375	0.299	0.343	0.290	0.342	0.379	0.419
		192	0.314	0.361	0.348	<u>0.362</u>	0.361	0.381	0.323	0.364	<u>0.317</u>	0.365	0.363	0.380	0.363	0.370	0.327	0.387	0.335	0.365	0.332	0.369	0.426	0.441
		336	<u>0.349</u>	0.380	0.376	0.391	0.393	0.402	0.350	<u>0.381</u>	0.346	0.382	0.395	0.403	0.392	0.390	0.410	0.411	0.369	0.386	0.366	0.392	0.445	0.459
		720	0.406	0.409	0.440	0.423	0.455	0.440	<u>0.411</u>	<u>0.411</u>	0.413	0.412	0.453	0.438	0.458	0.425	0.478	0.450	0.425	0.421	0.416	0.420	0.543	0.490
		Avg	0.336	0.372	0.340	0.378	0.381	0.394	<u>0.337</u>	0.374	0.338	<u>0.374</u>	0.384	0.395	0.383	0.384	0.400	0.406	0.357	0.378	0.351	0.380	0.448	0.452
ETTm2		96	<u>0.163</u>	0.253	0.170	0.245	0.176	0.258	0.165	0.258	0.162	0.253	0.177	0.259	0.169	<u>0.248</u>	0.187	0.267	0.167	0.269	0.166	0.255	0.203	0.287
		192	0.217	0.293	0.229	<u>0.291</u>	0.239	0.300	0.221	0.295	<u>0.218</u>	0.293	0.243	0.301	0.234	0.292	0.249	0.309	0.224	0.303	0.223	0.296	0.269	0.328
		336	<u>0.271</u>	0.330	0.303	0.343	0.297	0.338	0.271	0.331	0.270	0.328	0.302	0.340	0.294	0.339	0.321	0.351	0.281	0.342	0.274	<u>0.329</u>	0.325	0.366
		720	<u>0.348</u>	0.379	0.373	0.399	0.393	0.396	0.347	0.381	0.349	0.379	0.397	0.396	0.390	0.388	0.408	0.403	0.397	0.421	0.362	<u>0.385</u>	0.421	0.415
		Avg	0.250	<u>0.314</u>	0.269	0.320	0.276	0.323	0.251	0.316	0.250	0.313	0.279	0.324	0.272	0.317	0.291	0.333	0.267	0.333	0.255	0.315	0.304	0.349
Weather		96	0.146	0.196	0.155	0.205	0.156	0.204	0.145	<u>0.195</u>	<u>0.145</u>	0.195	0.163	0.207	0.150	0.188	0.172	0.220	0.176	0.237	0.149	0.198	0.217	0.296
		192	<u>0.192</u>	<u>0.241</u>	0.201	0.245	0.209	0.249	0.191	0.241	0.192	0.241	0.211	0.251	0.202	0.238	0.219	0.261	0.220	0.282	0.194	0.241	0.276	0.336
		336	<u>0.242</u>	<u>0.279</u>	0.237	0.265	0.264	0.290	0.243	0.282	0.243	0.281	0.267	0.292	0.260	0.282	0.280	0.306	0.265	0.319	0.245	0.282	0.339	0.380
		720	0.307	0.328	0.312	0.334	0.343	0.342	0.315	0.336	<u>0.307</u>	0.328	0.343	0.341	0.343	0.353	0.365	0.359	0.333	0.362	0.314	0.334	0.403	0.428
		Avg	0.221	0.261	0.226	0.262	0.243	0.271	0.223	0.263	<u>0.222</u>	<u>0.261</u>	0.246	0.272	0.239	0.261	0.259	0.287	0.248	0.300	0.225	0.263	0.309	0.360
Electricity		96	0.130	0.227	0.135	0.222	0.147	0.243	<u>0.128</u>	<u>0.221</u>	0.127	0.221	0.147	0.241	0.141	0.233	0.168	0.272	0.197	0.282	0.190	0.296	0.193	0.308
		192	0.144	0.241	0.147	<u>0.235</u>	0.157	0.251	<u>0.145</u>	<u>0.227</u>	0.145	0.240	0.165	0.258	0.160	0.250	0.184	0.298	0.196	0.285	0.199	0.304	0.201	0.315
		336	<u>0.161</u>	0.259	0.164	0.245	0.174	0.269	0.165	<u>0.252</u>	0.160	0.256	0.177	0.273	0.173	0.263	0.198	0.300	0.209	0.301	0.217	0.319	0.214	0.329
		720	0.193	0.289	0.212	0.310	0.206	0.296	0.194	0.295	0.195	<u>0.288</u>	0.213	0.304	0.197	0.284	0.220	0.320	0.245	0.333	0.258	0.352	0.246	0.355
		Avg	0.157	0.254	0.165	0.253	0.171	0.265	<u>0.158</u>	0.249	0.157	<u>0.251</u>	0.175	0.269	0.168	0.258	0.193	0.295	0.212	0.300	0.216	0.318	0.214	0.327
Traffic		96	<u>0.353</u>	<u>0.242</u>	0.392	0.253	0.455	0.298	0.358	0.244	0.352	0.240	0.406	0.277	0.419	0.269	0.593	0.321	0.410	0.282	0.360	0.249	0.587	0.366
		192	<u>0.364</u>	<u>0.249</u>	0.402	0.258	0.470	0.316	0.375	0.255	0.364	0.249	0.426	0.290	0.443	0.276	0.617	0.336	0.423	0.287	0.375	0.256	0.604	0.373
		336	0.377	0.257	0.428	0.263	0.479	0.316	0.386	0.272	<u>0.378</u>	<u>0.257</u>	0.432	0.281	0.460	0.283	0.629	0.336	0.436	0.296	0.392	0.264	0.621	0.383
		720	0.421	0.281	0.441	<u>0.282</u>	0.523	0.328	0.427	0.290	<u>0.422</u>	0.283	0.463	0.300	0.490	0.299	0.640	0.350	0.466	0.315	0.432	0.286	0.626	0.382
		Avg	0.379	<u>0.258</u>	0.416	0.264	0.482	0.315	0.387	0.265	<u>0.379</u>	0.257	0.461	0.287	0.453	0.282	0.620	0.336	0.434	0.295	0.390	0.269	0.609	0.376

Table 2: Ablation study for GFMixer. “w/o Acc” removes the acceleration gate in TGB; “w/o Trend” removes the trend gate; “w/o TGB” removes the entire TGB path. Best results are in bold and second-best in underlined.

Datasets	ETTh1		ETTh2		ETTm1		Weather	
Method	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
GFMixer	0.398	0.422	0.327	0.378	0.336	0.372	0.221	0.261
w/o TGB	0.410	0.428	<u>0.333</u>	0.382	0.338	<u>0.374</u>	<u>0.226</u>	<u>0.262</u>
w/o Acc	0.408	0.428	0.334	0.384	<u>0.337</u>	0.378	0.227	0.271
w/o Trend	<u>0.404</u>	<u>0.427</u>	0.332	<u>0.379</u>	0.340	0.375	0.229	0.269
w/o FAB	0.409	0.428	0.335	0.385	0.341	0.375	0.229	0.282

trend signal (first-order gradient) and excludes the acceleration signal (second-order gradient); and (iii) **w/o Trend**, which retains only the acceleration signal. The results in Table 2 show that removing the entire TGB (w/o TGB) consistently leads to performance drops across all datasets, confirming that TGB provides tangible benefits. Interestingly, retaining either the trend (w/o Acc) or acceleration (w/o Trend) signals alone sometimes yields results comparable to those of GFMixer. However, on datasets like ETTh2 and Weather,

these one-sided gradients perform worse than the version without TGB. This suggests that isolated non-task-adapted gradient features may introduce noise rather than informative patterns. These results emphasize the importance of adaptively incorporating gradient signals based on the Task-Adaptive Gating mechanism.

Effect of Fourier-Aware Attention Block. We further examine the contribution of the Fourier-Aware Attention Block (FAB) by replacing it with a standard attention mechanism (**w/o FAB**). As shown in Table 2, removing FAB consistently results in degraded performance across all datasets, with particularly large drops on ETTh1 and ETTh2. Given that these datasets exhibit strong periodic patterns, this degradation is especially significant, suggesting that FAB plays a crucial role in preserving frequency-specific structures essential for accurate long-term forecasting. These results confirm that FAB enhances the model’s ability to capture and retain critical frequency components, mitigating spectral degradation during deep representation learning. Importantly, while our main experiments focus on datasets with visible periodicity, GFMixer imposes no architectural assumptions specific to periodic signals. The FAB module serves as a general-purpose attention mechanism applicable to arbitrary temporal sequences. GFMixer’s consistent

Table 3: Plug-and-Play test for GFMixer. A * denotes backbones augmented with GFMixer. Within each backbone group, values are compared pairwise, and the better result is marked in bold.

Model Datasets	Transformer*		Transformer		Autoformer*		Autoformer		DLinear*		DLinear	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.926	0.773	0.949	0.768	0.501	0.489	0.547	0.507	0.460	0.457	0.462	0.458
ETTh2	3.968	1.598	4.257	1.639	0.467	0.468	0.580	0.499	0.560	0.518	0.562	0.519
ETTm1	0.840	0.678	0.933	0.720	0.544	0.498	0.563	0.514	0.435	0.474	0.423	0.464
ETTm2	1.537	0.881	1.388	0.848	0.357	0.373	0.385	0.379	0.340	0.391	0.344	0.395

performance improvements on datasets with weaker periodicity, such as Weather, demonstrate its ability to generalize across both periodic and non-periodic time series forecasting tasks.

5.3 Top-K Hyperparameter and TGB Effectiveness

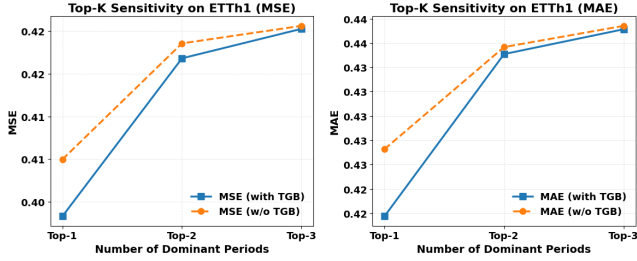


Figure 3: Sensitivity analysis of Top- k for GFMixer on the ETTh1 dataset with an averaged prediction horizon. We compare forecasting performance with and without the Temporal Gradient Block (TGB) under varying numbers of retained frequency components.

We study the sensitivity of frequency-domain forecasters to the number of retained frequencies (Top- k) and its interaction with TGB, following prior practice [9, 17]. As shown in Figure 3, on ETTh1, a small k leaves more low-amplitude residual signals unmodeled by the frequency path, so TGB delivers sizable gains by compensating for those cues; as k increases and the spectral path captures more content, the gap narrows. Beyond an optimal k , adding additional low-amplitude components introduces spectral noise, slightly degrading accuracy and reducing TGB’s marginal benefit. These results support our design: TGB acts as a residual compensator for patterns missed by the dominant periodic structure, while a modest k strikes a better signal-to-noise trade-off.

5.4 Plug-and-Play Transferability across Heterogeneous Backbones

Beyond achieving strong standalone results, a key promise of our design is composability: TGB, FAB, and GAB should act as drop-in modules that consistently improve diverse forecasters without bespoke re-engineering. This section quantitatively validates

such plug-and-play transferability. We select representative models spanning linear, decomposition, and frequency-centric families: Transformer [14], Autoformer [18] and DLinear [22]. Experiments follow the same benchmarks and official splits used throughout the paper (ETTh1/2, ETTm1/2) with prediction horizons $H \in \{96, 192, 336, 720\}$ ⁴.

To ensure a fair comparison, we adopt a minimal-intrusion policy. For Attention-based backbones (e.g., Transformer and Autoformer), we replace the native attention with FAB. Meanwhile, the TGB and GAB modules compute trend and acceleration evidence in parallel and inject them as a residual correction, ensuring that the backbone’s forward and training loops remain intact. For linear backbones like DLinear, we apply TGB and GAB only, as there is no attention mechanism to swap. As Table 3 shows, our modules can be connected to different backbone networks with minimal modifications, achieving overall steady improvements in MSE and MAE, particularly on Autoformer and Transformer. DLinear models also yield consistent benefits, validating our core proposition of “decoupling and plug-and-play.”

6 Conclusions

We present GFMixer, a dual-path, decoupled framework for long-term multivariate time series forecasting. GFMixer tackles two structural issues in frequency-domain modeling: frequency bias and spectral degradation. The Temporal Gradient Block (TGB) injects temporal-geometric residuals along the time axis, while the Fourier-Aware Attention Block (FAB) parameterizes attention over multi-frequency representations to preserve spectral integrity. Across diverse benchmarks, GFMixer consistently surpasses strong baselines and delivers plug-and-play gains on mainstream backbones; ablations confirm that TGB and FAB contribute independently and synergistically.

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⁴All plug-and-play experiments use the official implementations and settings from <https://github.com/thuml/Time-Series-Library>.

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A Theoretical Guarantee of GFMixer Framework

This section provides theoretical guarantees tailored to GFMixer:

- (1) **Frequency-bias gap**: We lower-bound the irrecoverable loss for top- k spectral truncation, explaining why weak components are lost;
- (2) **Stability-gain tension**: We show that stable in-backbone amplification limits recoverable transient energy to $O(k/\Omega)$;
- (3) **Spectral degradation**: We prove that standard layers suffer from cross-frequency leakage and harmonic dilution;
- (4) **Preservation condition**: We demonstrate that the FAB's decoupled residual path preserves backbone periodic energy with a bounded correction.

A.1 Frequency Bias Gap

A.1.1 Notation and Transient Class definition. Let $x \in \mathbb{R}^N$ and $F \in \mathbb{C}^{N \times N}$ be the unitary DFT ($F^*F = I$), $\hat{x} = Fx$ (Parseval: $\|x\|_2 = \|\hat{x}\|_2$). Decompose

$$x = p + r + \varepsilon, \quad \hat{p} = Fp, \quad \hat{r} = Fr, \quad \hat{\varepsilon} = F\varepsilon,$$

where p is a high-energy (few-line) backbone, r a low-energy transient, and ε noise. For $\Omega_k \subseteq \{0, \dots, N-1\}$, $|\Omega_k| = k$, define $\mathcal{V}_k \triangleq \{z \in \mathbb{C}^N : \text{supp}(z) \subseteq \Omega_k\}$ and the orthogonal projector $P_k \triangleq F^* \Pi_{\Omega_k} F$.

Definition A.1 (Class $\mathcal{R}(s, \Omega, \eta)$). For integers $s, \Omega \geq 1$ and $\eta \geq 1$, we say $r \in \mathcal{R}(s, \Omega, \eta)$ if

- (1) (Time-local) $|\text{supp}(r)| \leq s$;
- (2) (Spectral spread) with $S \triangleq \text{supp}(\hat{r})$, we have $|S| \geq \Omega$;
- (3) (Weak flatness) $\max_{\omega \in S} |\hat{r}_\omega| \leq \eta \|r\|_2 / \sqrt{|S|} = \eta \|r\|_2 / \sqrt{\Omega}$.

REMARK A.1 (UNCERTAINTY). By the Donoho–Stark uncertainty principle, $|\text{supp}(r)| \cdot |\text{supp}(\hat{r})| \geq N$, hence short transients imply broad spectra: $|S| \geq N/s$.

A.1.2 Frequency Bias.

LEMMA A.2 (RETAINED TRANSIENT ENERGY UNDER k LINES). If $r \in \mathcal{R}(s, \Omega, \eta)$, then for any Ω_k with $|\Omega_k| = k$,

$$\|P_k r\|_2^2 = \sum_{\omega \in \Omega_k} |\hat{r}_\omega|^2 \leq k \max_{\omega \in S} |\hat{r}_\omega|^2 \leq \eta^2 \frac{k}{\Omega} \|r\|_2^2.$$

THEOREM A.3 (UNAVOIDABLE LOSS). For $r \in \mathcal{R}(s, \Omega, \eta)$ and any k -line projector P_k ,

$$\|(I - P_k)r\|_2^2 \geq \max \left\{ 0, \left(1 - \eta^2 \frac{k}{\Omega} \right) \|r\|_2^2 \right\}.$$

In particular, if $\Omega \gg k$ (typical short-transient regime), a constant fraction $1 - O(k/\Omega)$ of transient energy is inevitably discarded.

REMARK A.2. *The bound is nontrivial when $\eta^2 k/\Omega < 1$; otherwise the trivial lower bound 0 applies.*

A.1.3 *Risk Floor When Only $P_k x$ Is Observable.* Let $y = \langle \beta, x \rangle + \xi$ with $\mathbb{E}[\xi | x] = 0$ and $\text{Var}(\xi | x) = \sigma^2$. Write $\beta = \beta_{\parallel} + \beta_{\perp}$ with $\beta_{\parallel} \in \mathcal{V}_k$ and $\beta_{\perp} \perp \mathcal{V}_k$, and set $Z = P_k x$.

PROPOSITION A.4 (EXACT EXCESS BAYES RISK).

$$\begin{aligned} \text{ExcessMSE} &\triangleq \inf_{\phi} \mathbb{E}[(y - \phi(Z))^2] - \inf_{\psi} \mathbb{E}[(y - \psi(x))^2] \\ &= \mathbb{E}[\text{Var}(\langle \beta_{\perp}, (I - P_k)x \rangle | Z)]. \end{aligned}$$

ASSUMPTION A.1 (ALIGNMENT & CONDITIONAL ORTHOGONALITY). Let $u = r/\|r\|_2$. Define the alignment coefficient

$$\alpha = \frac{|\langle \beta_{\perp}, r \rangle|}{\|\beta_{\perp}\|_2 \|r\|_2} \in [0, 1].$$

Write $\beta_{\perp} = \alpha \|\beta_{\perp}\|_2 u + \beta_{\perp}^{\perp}$ with $\langle u, \beta_{\perp}^{\perp} \rangle = 0$. Assume $\text{Cov}(\langle \beta_{\perp}^{\perp}, (I - P_k)r \rangle, \langle u, (I - P_k)r \rangle | Z) = 0$.

LEMMA A.5 (GRADIENT IDENTITY). With $b = (I - P_k)r$, $\langle u, b \rangle = \|b\|_2^2/\|r\|_2$, hence $\langle u, b \rangle^2 = \|b\|_2^4/\|r\|_2^2$.

THEOREM A.6 (RISK LOWER BOUND). Under Assumption A.1,

$$\begin{aligned} \text{ExcessMSE} &\geq \|\beta_{\perp}\|_2^2 \alpha^2 \mathbb{E} \left[\frac{\|(I - P_k)r\|_2^4}{\|r\|_2^2} \right] \\ &\geq \|\beta_{\perp}\|_2^2 \alpha^2 \left(1 - \eta^2 \frac{k}{\Omega} \right)^2 \mathbb{E} \|r\|_2^2. \end{aligned}$$

REMARK A.3. If $\beta_{\perp} \parallel r$ ($\alpha = 1$), the first inequality is tight in the variance step, yielding $\text{ExcessMSE} \geq \|\beta_{\perp}\|_2^2 (1 - \eta^2 k/\Omega)^2 \mathbb{E} \|r\|_2^2$.

A.1.4 *Amplification Instability.* Let $A_w = F^* \text{diag}(w) F$. Set $w_{\omega} = \gamma > 1$ for $\omega \in S$ and $w_{\omega} \approx 1$ otherwise.

PROPOSITION A.7 (NOISE AMPLIFICATION & CONDITIONING). First, $\|A_w\|_2 = \max_{\omega} |w_{\omega}| \geq \gamma$. If $\min_{\omega} |w_{\omega}| \geq c > 0$ and $\exists w_{\omega} = 1$, then

$$\kappa(A_w) = \|A_w\|_2 \|A_w^{-1}\|_2 = \frac{\max_{\omega} |w_{\omega}|}{\min_{\omega} |w_{\omega}|} \geq \frac{\gamma}{c} \geq \gamma.$$

If $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, then $\mathbb{E} \|A_w \varepsilon\|_2^2 = \sigma^2 \text{tr}(A_w A_w^*) = \sigma^2 \sum_{\omega} |w_{\omega}|^2 \geq \sigma^2 |S| \gamma^2$.

PROPOSITION A.8 (BOUNDED GAIN UNDER STABILITY). If $\|A_w\|_2 \leq C$ and $r \in \mathcal{R}(s, \Omega, \eta)$, then for any Ω_k ,

$$\|P_k A_w r\|_2^2 \leq C^2 \sum_{\omega \in \Omega_k} |\widehat{r}_{\omega}|^2 \leq C^2 \eta^2 \frac{k}{\Omega} \|r\|_2^2.$$

Thus, keeping the operator stable cannot overcome the $\widetilde{O}(k/\Omega)$ limitation in Lemma A.2.

A.1.5 *Algorithm Description.*

- (1) **Structural loss:** Any k -line retention discards a constant fraction of spread transient energy (Theorem A.3);
- (2) **Risk floor:** Under mild alignment, this loss induces a non-vanishing prediction error when only $P_k x$ is visible (Theorem A.6);
- (3) **Instability of forcing-up:** Large low-band gains inflate noise and condition number (Proposition A.7);
- (4) **Bounded gain:** Stability constraints cap recoverable transient energy at $\widetilde{O}(k/\Omega)$ (Proposition A.8).

These points justify a decoupled design: preserve the backbone along a frequency-aware path, and extract transients on a separate geometric path and inject residually.

A.2 Spectral Degradation in Deep Stacks

A.2.1 *Setup, Units, and Operators.*

Sampling & frequency units. $x[t] \in \mathbb{R}^N$, $t = 0, \dots, T-1$, sampling rate f_s (Hz); normalized angular frequency $\omega \in [-\pi, \pi)$ with $\omega = 2\pi f/f_s$. DC ($\omega = 0$) is treated separately and never mapped to a finite period.

Period-aware patching with padding. For $\omega_i \in (0, \pi]$, let $p_i = \text{round}(2\pi/\omega_i) \in \mathbb{Z}_{\geq 1}$, $L_i = \lceil T/p_i \rceil$, $T'_i = L_i p_i$. Zero-pad to T'_i , and define $Z^{(i)}[n](\tau) = x[np_i + \tau] \in \mathbb{R}^N$ for $n = 0, \dots, L_i - 1$ and $\tau = 0, \dots, p_i - 1$ (padding when $np_i + \tau \geq T$). Let F_{p_i} be the unitary DFT on $\mathbb{R}^{p_i}$, $D = F_{p_i} \otimes I_N$, S the cyclic shift on $\mathbb{R}^{p_i}$. Denote Π_{p_i} the projector onto the p_i -periodic subspace and Π_{ℓ} the projector onto intra-patch DFT bin ℓ .

A.2.2 *Quantized-Period Drift and DFT Leakage.*

Exact phase drift. For $p[t] = a \cos(\omega t + \phi)$, define the quantization misfit

$$\delta = \omega p_i - 2\pi k_i, \quad k_i = \arg \min_{k \in \mathbb{Z}} |\omega p_i - 2\pi k|.$$

If $p_i = \text{round}(2\pi/\omega)$ (so $k_i = 1$), then $\delta = \omega(p_i - 2\pi/\omega)$ and hence $|\delta| \leq \omega/2$. Then for any τ , the drift between adjacent patches is:

$$p[(n+1)p_i + \tau] - p[np_i + \tau] = -2a \sin(\omega\tau + \phi + (n + \frac{1}{2})\delta) \sin\left(\frac{\delta}{2}\right).$$

Hence for fixed τ , the patch-indexed sequence

$$p[np_i + \tau] = a \cos(\omega\tau + \phi + n\delta)$$

has correlation $\cos(m\delta)$ at lag m .

Finite-window leakage via Fejér kernel. Let $x_{\tau} = a \cos(\omega\tau + \phi)$, $\tau = 0, \dots, p_i - 1$, with unitary DFT $\widehat{x}[m]$. Let $\Delta = \omega - 2\pi m^*/p_i$ be the offset to the nearest bin m^* , and define the Fejér kernel

$$\text{Fej}_{p_i}(\Delta) = \frac{1}{p_i} \left(\frac{\sin(\frac{p_i \Delta}{2})}{\sin(\frac{\Delta}{2})} \right)^2.$$

Then there exist absolute constants $c_1, c_2 > 0$ such that

$$|\widehat{x}[m^*]|^2 \leq c_2 a^2 \text{Fej}_{p_i}(\Delta), \quad \frac{\sum_{m \neq m^*} |\widehat{x}[m]|^2}{\sum_m |\widehat{x}[m]|^2} \geq 1 - c_1 \text{Fej}_{p_i}(\Delta).$$

Thus, for fixed nonzero Δ , a fixed fraction of energy leaks outside m^* (does not vanish with p_i).

A.2.3 *Additive Positional Encoding (APE).*

LEMMA A.9 (PERIODIC PROJECTION IDENTITY). For any patch Z , APE P , and projectors $(\Pi_{p_i}, \Pi_{p_i}^{\perp})$,

$$\begin{aligned} \|\Pi_{p_i}(Z + P)\|_F^2 &= \|\Pi_{p_i} Z\|_F^2 + 2\langle \Pi_{p_i} Z, \Pi_{p_i} P \rangle_F + \|\Pi_{p_i} P\|_F^2, \\ \|\Pi_{p_i}^{\perp}(Z + P)\|_F^2 &= \|\Pi_{p_i}^{\perp} Z\|_F^2 + 2\langle \Pi_{p_i}^{\perp} Z, \Pi_{p_i}^{\perp} P \rangle_F + \|\Pi_{p_i}^{\perp} P\|_F^2. \end{aligned}$$

PROPOSITION A.10 (HARMLESS APE $\Leftrightarrow$ PERIODIC & PATCH-STATIONARY). If $\Pi_{p_i}^{\perp} P[n] \neq 0$ for some n , then (unless exact cancellation occurs) the nonperiodic energy increases. Moreover, cross-patch similarity degrades unless $P[n] \equiv P$ is n -stationary and $P \in \mathcal{V}_{p_i}$ (period p_i). Generic learned APE thus injects out-of-period energy and drift.

Table 4: Dataset descriptions. *Samples* are organized in (Train/Validation/Test).

TASKS	DATASETS	CHANNELS	SERIES LENGTH	SAMPLES	INFORMATION	FREQUENCY
Long-term	ETTH1,ETTH2	7	{96,192,336,720}	8,545/2,881/2,881	Electricity	Hourly
	ETTM1,ETTM2	7	{96,192,336,720}	34,465/11,521/11,521	Electricity	15 Mins
	Weather	21	{96,192,336,720}	36,792/5,271/10,540	Weather	10 Mins
	Exchange	8	{96,192,336,720}	5,120/665/1,422	Exchange rate	Daily
	Electricity	321	{96,192,336,720}	18,317/2,633/5,261	Electricity	Hourly
	Traffic	862	{96,192,336,720}	12,185/1,757/3,509	Transportation	Hourly
	Solar-Energy	137	{96,192,336,720}	36,601/5,161/10,417	Energy	10 Mins

A.2.4 Linear Maps: Frequency Preservation $\Leftrightarrow$ Commutation. Flatten $z \in \mathbb{R}^{p_i N}$ and apply a linear map $y = Wz$ with $W \in \mathbb{R}^{p_i N \times p_i N}$.

THEOREM A.11 (BLOCK-CIRCULANT/COMMUTATION CRITERION).

The following are equivalent:

- (1) For every ℓ , $\Pi_\ell Dz = Dz \implies \Pi_\ell(DWD^{-1})Dz = (DWD^{-1})Dz$;
- (2) $(S \otimes I_N)W = W(S \otimes I_N)$ (commutes with cyclic shift);
- (3) $W = D^{-1}\tilde{W}D$ with block-diagonal $\tilde{W}$ (blocks indexed by ℓ).

If $[W, D] \triangleq WD - DW \neq 0$, then for some ℓ ,

$$\sup_{\|z\|_2=1} \|(I - \Pi_\ell)DWD^{-1}\Pi_\ell Dz\|_2 \geq c \| [W, D] \|_2$$

for an absolute $c > 0$.

REMARK A.4. Patchwise flatten $\rightarrow$ FC violates Theorem A.11 and must mix bins unless designed block-circulant (LTI on the patch axis).

A.2.5 Memoryless Nonlinearities: Harmonics & Intermodulation. Let analytic $\phi(u) = \sum_{m \geq 0} a_m u^m$; piecewise-linear cases follow by local polynomialization on bounded-energy sets.

THEOREM A.12 (SPECTRAL CONVOLUTION). For the input signal

$$x(\tau) = \sum_{j=1}^J A_j \cos(\omega_j \tau + \varphi_j),$$

the DFT of $\phi(x)$ has support on Minkowski sums $\sum_j s_j \omega_j$ ($s_j \in \mathbb{Z}$) modulo 2π . Let m_j be the nearest DFT bin index to ω_j . For $J = 2$ and non-affine ϕ :

$$\|(I - \Pi_{m_1} - \Pi_{m_2})\widehat{\phi(x)}\|_2^2 \geq c_2 |a_2|^2 A_1^2 A_2^2 + c_3 |a_3|^2 (A_1^6 + A_2^6) + \dots$$

where coefficients $c_m > 0$ depend on the DFT normalization and window length.

REMARK A.5 (SMALL-SIGNAL EXPLICIT BOUND). If $\phi(u) = \alpha u + \frac{\beta}{2} u^2 + o(u^2)$ and amplitudes are small, the line at $\omega_1 \pm \omega_2$ has magnitude $\geq c|\beta|A_1 A_2$ for some $c > 0$.

A.3 One-Layer Quantitative Degradation

Consider one layer $z \xrightarrow{\text{APE}} z + P \xrightarrow{W} W(z + P) \xrightarrow{\phi} \phi(W(z + P))$. Assume: (i) $\|\Pi_{p_i}^\perp P\|_F > 0$ or P is non-stationary in n ; (ii) $[W, D] \neq 0$; (iii) ϕ is non-affine with some $a_m \neq 0$ ($m \geq 2$). Define the periodic energy ratio $R(u) = \|\Pi_{p_i} u\|_F^2 / \|u\|_F^2$.

THEOREM A.13 (STRICT DROP PER LAYER). There exist absolute constants $\rho_{\text{ape}}, \rho_{\text{lin}}, \rho_{\text{nl}} > 0$ such that for all $z \neq 0$,

$$R(\phi(W(z + P))) \leq R(z) - \rho_{\text{ape}} - \rho_{\text{lin}} - \rho_{\text{nl}} + \mathcal{O}(p_i^{-1}).$$

The reduction terms satisfy:

$$\begin{aligned} \rho_{\text{ape}} &\geq c_{\text{ape}} \frac{\|\Pi_{p_i}^\perp P\|_F^2}{\|z\|_2^2 + \|P\|_F^2}, \\ \rho_{\text{lin}} &\geq c_{\text{lin}} \frac{\|[W, D]\|_2^2}{1 + \|W\|_2^2}, \\ \rho_{\text{nl}} &\geq c_{\text{nl}} \sum_{m \geq 2} |a_m|^2 \Theta_m(z), \end{aligned}$$

where $c_{\text{ape}}, c_{\text{lin}}, c_{\text{nl}} > 0$ are universal constants, and $\Theta_m(\cdot) \geq 0$ captures the m -th order intermodulation energy created by ϕ on $W(z + P)$.

REMARK A.6 (NO-DEGRADATION IFF). Equality (no degradation) holds iff P is n -stationary and $P \in \mathcal{V}_{p_i}$, $[W, D] = 0$, and ϕ is affine.

B Dataset Description

B.1 Long-Term Datasets

- **ETT (4 subsets)**⁵ contains oil temperature and power load series collected by electricity transformers from July 2016 to July 2018, recorded at 15-minute or hourly intervals.
- **Weather**⁶ includes meteorological series with 21 indicators collected every 10 minutes by the Max Planck Biogeochemistry Institute in 2020.
- **Electricity**⁷ records hourly electricity consumption of 321 clients from 2012 to 2014.
- **Traffic** measures hourly road occupancy rates from 862 sensors deployed in the San Francisco Bay Area freeways between January 2015 and December 2016.

⁵<https://github.com/zhouhaoyi/Informer2020>

⁶<https://www.bgc-jena.mpg.de/wetter/>

⁷<http://pems.dot.ca.gov/>